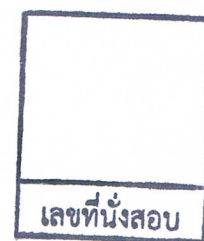


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Subject: 392132 Physics II

Date: 23 December 2013

Year: 2013

Section: 15 - 16

Time: 13.00 - 15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken inside the examination room.
 3. Textbooks are **NOT** allowed.
 4. Express your answer in English. Other languages will be discarded.
 5. A scientific calculator is allowed.
 6. Electronic communication devices are **NOT** allowed in the examination room.
 7. The examination has 6 pages (including this page), 9 questions and a total score of 50 points.
 8. Write all your answers on this exam sheet.
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1. A ball is projected at a speed of 50.0 m/s at an angle of 35.0° with the horizontal

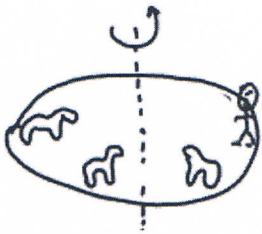
- (a) Maximum high (2 points)
- (b) Time at maximum high (2 points)
- (c) Velocity at maximum high (2 points)
- (d) Distance from start point to maximum high (3 points)
- (e) How far does a ball hit the ground? (2 points)

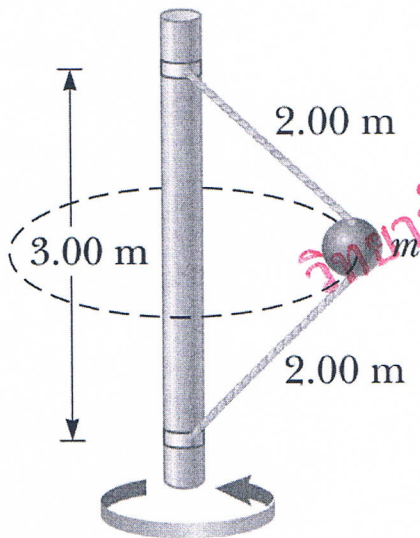
วิทยาลัยเทคโนโลยีอุตสาหกรรม

4. A 40.0-kg child stands at the rim of a merry-go-round of radius 2.00 m, rotating with an angular speed of 3.00 rad/s.

(a) What is the child's centripetal acceleration? (2 points)

(b) What is the minimum force between her feet and the floor of the carousel that is required to keep her in the circular path? (2 points)





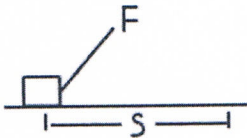
5. A 4.00-kg object is attached to a vertical rod by two strings as shown in Figure. The object rotates in a horizontal circle at constant speed 6.00 m/s. Find the tension in

(a) the upper string. (3 points)

(b) the lower string. (3 points)

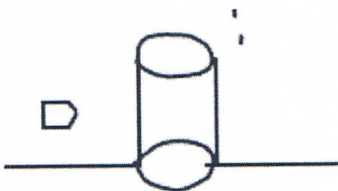
6. A sledge loaded with bricks has a total mass of 18.0 kg and is pulled at constant speed by a rope inclined at 20.0° above the horizontal. The sledge moves a distance of 20.0 m on a horizontal surface. The coefficient of kinetic friction between the sledge and surface is 0.500.

- (a) What is the tension in the rope? (2 points)
 (b) How much work is done by the rope on the sledge? (2 points)
 (c) What is the mechanical energy lost due to friction? (1 points)



7. A 7.80-g bullet moving at 550 m/s penetrates a tree trunk to a depth of 5.0 cm. (4 points)

- (a) Use work and energy considerations to find the average frictional force that stops the bullet.
 (b) Assuming the frictional force is constant, determine how much time elapses between the moment the bullet enters the tree and the moment it stops moving.

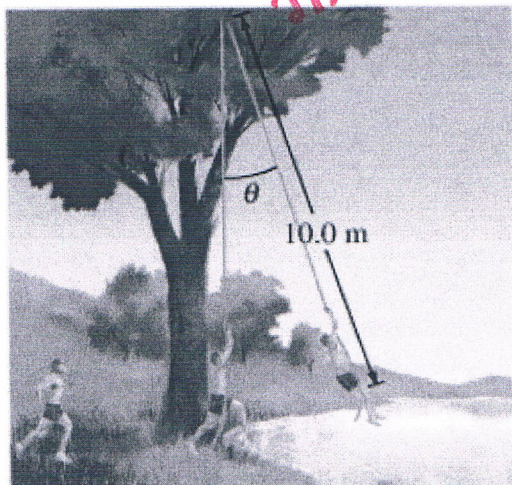


8. A 0.250-kg block along a horizontal track has a speed of 2.0 m/s immediately before colliding with a light spring of force constant 4.80 N/m located at the end of the track. What is the spring's maximum compression if the track is frictionless? (3 points)

9. A 65-kg student runs at 5.0 m/s, grabs a rope, and swings out over a lake. He releases the rope when his velocity is zero.

a) What is the angle θ when he releases the rope? (4 points)

b) What is the tension in the rope just before he releases it? (3 points)



Good Luck

Dr. Prakasit Changsuphan